Earth and Planetary Science Letters 554 (2021) 116656

Contents lists available at ScienceDirect

Earth and Planetary Science Letters

www.elsevier.com/locate/epsl

Chronology of serpentinization: Paleomagnetic evidence for 20 Ma serpentinization of the Oeyama ophiolite, Wakasa, Southwest Japan

Yo-ichiro Otofuji [a,*], Makoto Fujihara [b], Koji Uno [c]

[a] Institute of GeoHistory, Japan Geochronology Network, NPO, 1599 Susai, Akaiwa 701-2503, Japan
[b] Hiruzen Institute for Geology and Chronology, Co., Ltd. 2-5 Nakashima, Naka-ku, Okayama 703-8252, Japan
[c] Department of Earth Sciences, Okayama University, 3-1-1 Tsushimanaka, Kita-ku, Okayama 700-8530, Japan

ARTICLE INFO

Article history:
Received 6 June 2020
Received in revised form 20 October 2020
Accepted 29 October 2020
Available online 12 November 2020
Editor: A. Yin

Keywords:
paleomagnetism
rock magnetism
serpentine
serpentinization
ophiolite

ABSTRACT

Serpentinization of mantle peridotite generates magnetite that is able to acquire a remanent magnetization parallel to the ambient geomagnetic field. The paleomagnetic direction of serpentine provides insights as to the timing of serpentinization. This study examines paleomagnetic properties of serpentine in the Paleozoic Oeyama ophiolite in Wakasa (35.39°N, 134.39°E), Southwest Japan. Reversed polarity magnetizations with clockwise deflections from the south were observed in the serpentine in the high temperature component (HTC) magnetization with an unblocking temperature (T_{ub}) of ~580 °C. The mean reversed polarity direction was D = 226.0°, I = −37.8°, α_{95} = 5.9°, and n = 30. The remaining samples with an HTC of T_{ub} between 400-450 °C show the present-day geomagnetic field direction. Rock magnetism reveals that the reversed direction is carried by single-domain (SD) grains of magnetite, whereas multi-domain (MD) magnetite grains carry the present-day geomagnetic field direction. Magnetite grains originated from olivine during serpentinization grew to SD size and acquired the chemical remanent magnetizations during a reversed polarity interval. As serpentinization progresses, SD magnetite grains increased to MD magnetite grains and/or were subjected to maghemitization. The mean reversed direction yielded a paleomagnetic pole at 47.4°N, 230.3°E (A_{95} = 5.7°), which overlaps with the 20 Ma segment of the apparent polar wander path for SW Japan. We conclude that serpentinization occurred in the Oeyama ophiolite at approximately 20 Ma during the clockwise rotation of SW Japan.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction

Mantle peridotite is hydrated to form partially or completely serpentinized rocks. Serpentinization of mantle peridotite plays a crucial role in numerous natural processes at mid-ocean ridges, oceanic subduction zones and ophiolites on land; as such understanding serpentinization is important. Despite its geologic and tectonic importance, serpentinization dating is extremely difficult as minerals suitable for direct isotopic dating are not formed during serpentinization.

The rodingitization process occurred contemporaneously with serpentinization (Coulton et al., 1995). The serpentinization age was evaluated by dating rodingitized rocks, which commonly coexist with serpentinites (Coleman, 1967). Hydrothermal zircons precipitated from the hydrothermal fluid during rodingitization and were associated with serpentinization. Hydrothermal zircons yielded a U-Pb age of 291 Ma for the Changawuzi ophiolite in southwestern Tianshan (Li et al., 2010) and 400 Ma for the Sudetic ophiolite in Bohemia (Dubinska et al., 2004): researchers had assigned these zircon ages to the timing of serpentinization. However, as the source and mobility of Zr in the low-temperature fluid remain unclear during serpentinization, the geochronological interpretation of the thermal zircon may not always be definitive (Li et al., 2010). As such, many studies aim to determine reliable ages for serpentinization.

Paleomagnetism is an alternative method to estimate the age of serpentinization. Magnetite is a significant by-product of the serpentinization of mantle peridotite (e.g., O'Hanley, 1996). The transformation of olivine during serpentinization yields magnetite and serpentine. As a magnetite grain grows through the stable single-domain (SD) threshold size during serpentinization, it acquires a chemical remanent magnetization (CRM) as a natural remanent magnetization (NRM) parallel to the ambient geomagnetic field. The paleomagnetic pole calculated from the NRM of serpentine on a tectonic block represents the position at the time of serpentinization in its apparent polar wander path (APWP). The

* Corresponding author.
E-mail address: otofuji@kobe-u.ac.jp (Y.-i. Otofuji).

https://doi.org/10.1016/j.epsl.2020.116656
0012-821X/© 2020 Elsevier B.V. All rights reserved.

serpentinization-related remanence age was paleomagnetically estimated at 170 Ma for the Mirdits ophiolite in Albania (Maffione et al., 2013a), despite the sole use of inclination data.

The southwest (SW) Japan arc is an ideal place to compare the pole position of serpentine with its APWP. As SW Japan has been subjected to clockwise rotation of more than 40° since 100 Ma due to the opening of the Japan Sea, a large movement in paleomagnetic poles has been observed in SW Japan between the present-day and 100 Ma (Otofuji and Matsuda, 1987; Uno et al., 2017). Thus, it is reasonable to determine the timing of serpentinization in SW Japan at pre, syn or post-clockwise rotation.

The SW Japan arc bears ophiolitic complexes of various ages ranging from the early Paleozoic to the Cenozoic (Ishiwatari, 1991, 1994; Ishiwatari et al., 2003). The Oeyama ophiolite (Fig. 1) is the oldest complex with an age between 566–403 Ma; its main components are serpentinized peridotite (Kimura and Hayasaka, 2019). This study focuses on the serpentinite of the Wakasa ultramafic complexes (35.39°N, 134.39°E) in the Oeyama ophiolite. The paleomagnetic pole position of serpentinites at Wakasa was estimated, and the timing of serpentinization was subsequently examined.

2. Geological setting and sampling

SW Japan consists of sequentially developed accretionary complexes and associated ophiolites, which have developed oceanward since the middle Paleozoic (e.g., Moreno et al., 2016; Wallis et al., 2020). Ophiolites in SW Japan are arranged subparallel to the general elongation of the island arc; the age of ophiolites decreases towards the Pacific Ocean.

The Oeyama ophiolite, distributed along the north coast of the SW Japan arc, is the oldest ophiolite in SW Japan (Fig. 1). It is mainly composed of mainly lherzolite mantle peridotite, clinopyroxene-bearing harzburgite, and dunite, most of which are altered to serpentine (Ishiwatari and Tsujimori, 2003). Early Paleozoic ages have been assigned to the Oeyama ophiolite. The zircon $^{206}Pb/^{238}U$ ratio for gabbroic rocks yielded ages of 545 and 532 Ma; these are ascribed to the magmatic crystallization age during oceanic crustal formation (Kimura and Hayasaka, 2019). The K-Ar method for hornblende yielded ages between 469 and 403 Ma (Nishimura and Shibata, 1989; Tsujimori et al., 2000). These ages are interpreted as the timing of metasomatism during subduction (Kimura and Hayasaka, 2019). The Oeyama ophiolite may represent a supra-subduction zone ophiolite formed beneath an intra-oceanic arc (Tsujimori and Itaya, 1999; Ishiwatari and Tsujimori, 2003).

The Wakasa ultramafic complex is one of the exposed areas of the Oeyama ophiolite, and the hornblende indicates K-Ar ages between 444–469 Ma (Nishimura and Shibata, 1989). It is distributed in small areas along the Hosomi and Kurumino rivers (Uemura et al., 1979) (Fig. 1b). The Wakasa ultramafic complex is in fault contact with the Renge high-pressure/temperature (P/T) metamorphic belt, and the fault boundary tilts northeastward by 3°–4° (Uemura et al., 1979). The main components of the Wakasa ultramafic complex are dunite and its derived serpentinite.

The Wakasa ultramafic complex and the Renge high-P/T metamorphic belt are overlain by the lower Miocene Tottori Group and Miocene volcanic rocks (Matsumoto, 1986, 1989; Furuyama et al., 1997). The tilt of the Miocene strata provides paleohorizontal information for the Wakasa ultramafic complex later than 16 Ma. The mean tilt from nine localities in the Tottori Group was determined to be N37°W/10°E (Ishikawa et al., 2017), and the Miocene volcanic rocks provide N80°E/8°N in attitude (Uemura et al., 1979). Whole rock K-Ar ages of 2.60 ±0.08 Ma and 3.9 ± 1.5 Ma were assigned to the Miocene volcanic rocks (Uto et al., 1994).

Serpentinites of the Oeyama ophiolite are restricted to one locality 20 m wide along the Hosomi River in the Wakasa area (35.39°N, 134.39°E) (Fig. 1). The studied outcrop consists of two lithofacies, units A and B. Unit A is gray to black in color; although serpentinite is altered from dunite, the relict olivine has been preserved, as described by Uemura et al. (1979). In addition to serpentine and olivine, this unit also contains a minor amount of clinopyroxene and tremolite. Unit B is gray to green in color, and serpentinite is mainly altered from olivine clinopyroxenite. This unit mainly consists of serpentine, clinopyroxene, and tremolite. In each unit, the serpentinite is not constrained near the fault and appears to be a macroscopically homogeneous rock mass.

Forty-six block samples were collected from five sites in both units. Sampling site ③ was located in unit A, while sites ⑤, ⑦, ②, and ⑧ were in unit B. More than eight block samples were collected from each site with 1–2 m width. Samples were oriented in the field with a tripod-mounted magnetic compass. A declination correction of D = −9.1° was used, as the sun compass at sites ⑦ and ② provides a declination of −9.1° ± 0.8°.

3. Paleomagnetism

3.1. Laboratory procedures

Cylinders 2.5 cm in diameter were drilled from the 46 block samples and cut into 2.2 cm long specimens in the laboratory. The NRM of the specimens was measured at the laboratory of Okayama University using a spinner magnetometer (Natsuhara Giken SMM-85). Thermal demagnetization (ThD) was conducted using a thermal demagnetizer composed of a non-inductively wound electric furnace in three cylindrical μ-metal magnetic shields (Natsuhara Giken TDE-91C thermal demagnetizer). The magnetic field measured in the furnace was less than 20 nT. All specimens were subjected to progressive ThD in 12–16 heating steps up to a maximum temperature of 680 °C to detect the presence of hematite. ThD temperature intervals of 50 °C were used between 100 and 500 °C (nine steps), and 30 °C intervals were used from 500 to 680 °C (seven steps). An alternating field demagnetization (AFD) was also applied to the companion specimens for the 17 samples (Table 1). Specimens were subjected to progressive AFD in 10–15 steps at 2.5–20 mT intervals up to 100 mT, with a three-axis tumbler system in three cylindrical μ-metal magnetic shields (Natsuhara Giken DEM-93).

The demagnetization results were plotted on orthogonal vector diagrams (Zijderveld, 1967) and equal-area projections. Paleomagnetic directions were determined by principal component analysis based on non-anchored fits (Kirschvink, 1980). Site-mean and formation-mean directions were calculated using Fisherian statistics (Fisher, 1953).

3.2. Thermal demagnetization

One specimen was selected from each block sample with the exception for GY23. For this sample, two specimens were selected; as such, ThD was conducted on 47 samples. The 47 samples indicated initial NRM intensities ranging between 0.48 and 23 A/m (Appendix Table). Following the demagnetization of the low temperature component (LTC) at 250 °C, a high temperature component (HTC) with a maximum unblocking temperature (T_{ub}) of 580 °C appeared in 37 samples (Fig. 2). A single magnetic component with a maximum T_{ub} of 400-450 °C was observed in the remaining 10 samples.

Three kinds of NRMs were identified for the HTCs; SW, NE, and N remanences (Fig. 3, Table 1, and Appendix Table). Three directions and ThD behavior were used to classify 47 samples into five categories; SW_{SW}, SW_{NE}, SW_{ex}, NE, and N (Table 1).

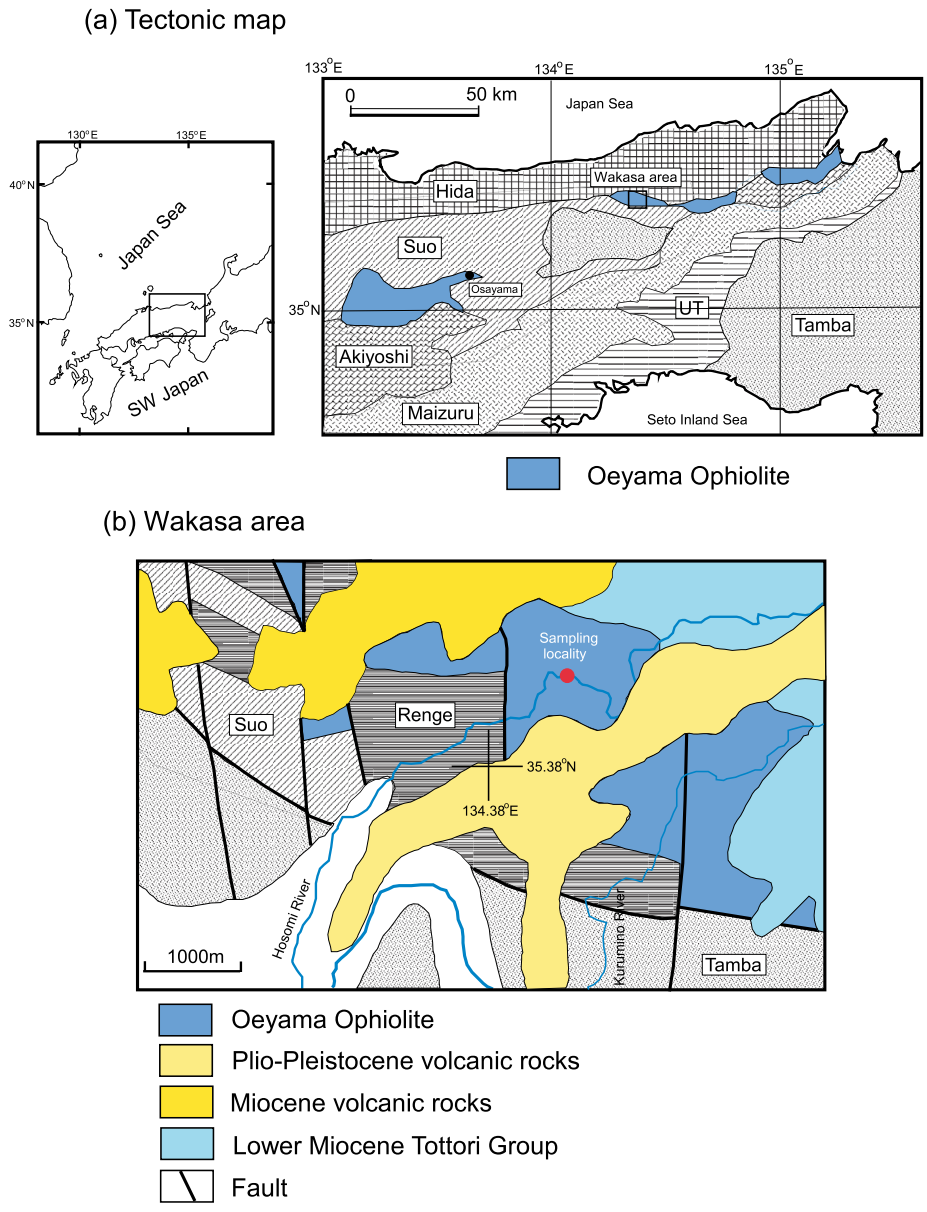


Fig. 1. Simplified tectonic and geological maps together with sketch map of sampling sites. (a) Tectonic framework of the study area showing the Oeyama Belt, and other Paleozoic and Mesozoic geotectonic units (Hida, Suo, Akiyoshi, Maizuru, Ultra-Tamba (UT), Tamba units) (Moreno et al., 2016); (b) simplified geological map of the Wakasa area (Uemura et al., 1979). Renge: Paleozoic geotectonic unit. Whole rock K-Ar ages of 2.60 ± 0.08 Ma and 3.9 ± 1.5 Ma were assigned to the Miocene volcanic rocks (Uto et al., 1994); (c) sketch map of sampling sites in the sampling locality. The numbers on the map correspond to sites. The studied outcrop was classified into two lithofacies, units A and B. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

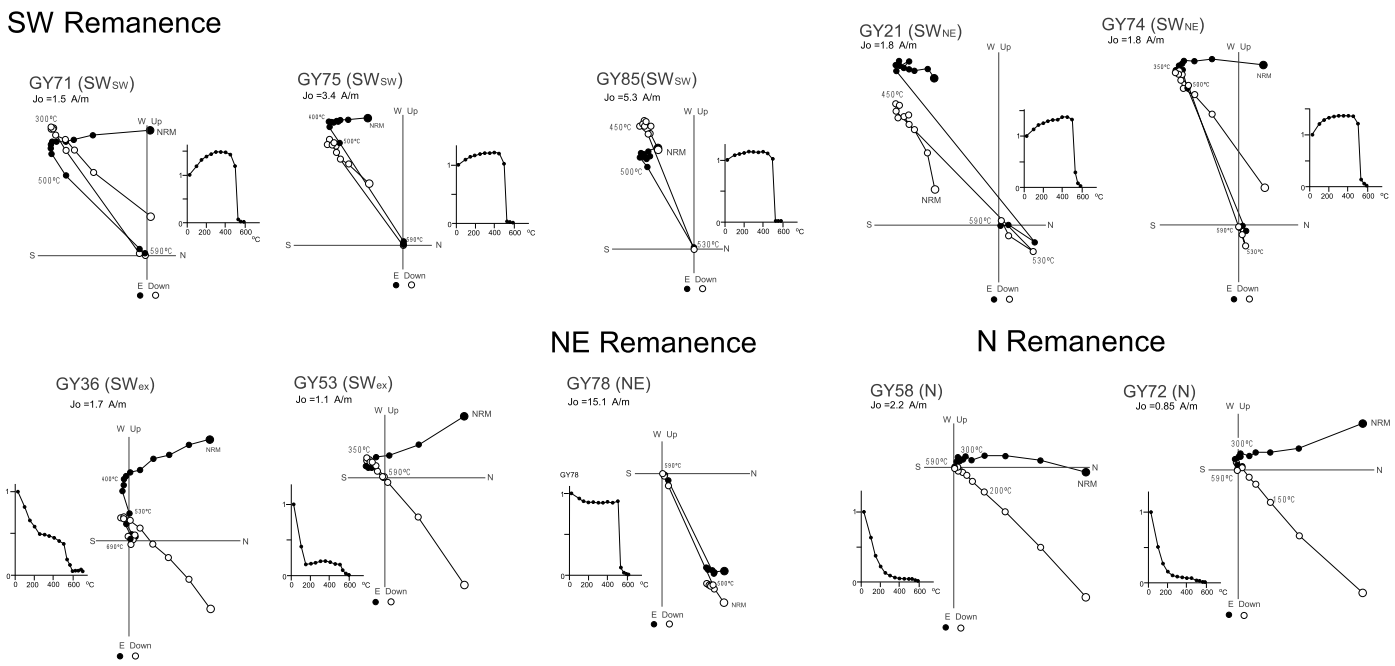


Fig. 2. Zijderveld plots and normalized demagnetization curves during stepwise thermal demagnetization for samples with SW, NE, and N remanences. Categories of thermal demagnetization behavior are attached to sample numbers. Open (solid) symbols in the Zijderveld plot indicate projection onto a vertical (horizontal) plane. All directions are plotted in geographic coordinates.

SW remanence (34 samples)

Reversed polarity directions with southwesterly declinations and upward inclinations were observed in 34 samples. Declinations varied between 170° and 259°, while inclinations varied between −7° and −50° (Fig. 3).

Stepwise ThD demonstrated that NRM intensity curves for all 34 samples showed a sharp intensity decrease at 500–530 °C (Fig. 2). A plateau or a gentle peak in the NRM intensity curves occurs up to 500–530 °C for 27 of the 34 samples. Only the SW remanence direction (category SW_{SW}) was observed in 19 of these 27 samples throughout the stepwise ThD, whereas a northeasterly direction opposite to the SW remanence direction appeared above 530 °C in eight samples (category SW_{NE}). The remaining seven samples (category SW_{ex}) showed swift and exponential decay in NRM intensity between room temperature and 250–350 °C before the SW remanence direction appeared.

NE remanence (3 samples)

Normal polarity directions with northeasterly declinations and downward inclinations appeared in three samples (category NE); that is, opposite to the SW remanence direction (Fig. 3). The stepwise ThD showed an abrupt decrease in NRM intensity at 500–530 °C as observed in SW remanence (Fig. 2). NE remanence directions were observed in the SW_{NE} category samples, although these directions were dispersed (Fig. 3).

N remanence (10 samples):

A northerly direction with a downward inclination was observed in 10 samples (category N). The HTCs were completely unblocked at 450 °C, and their NRM intensity curves were characterized by exponential decay during stepwise ThD (Fig. 2).

The remanence measurements indicated that the NRMs of serpentines were not homogeneous: they were variable within a volume of ∼1000 cm^3. Each sampling site was not consisting of samples with only one category (Table 1). Samples from sites ⑤ and ⑧ had three categories, while those from sites ②, ③ and ⑦ had four categories. Companion specimens within one block sample (GY23) had different categories; GY23-1 belonged to SW_{NE} whilst GY23-2 was a part of NE category.

3.3. NRM directions of serpentine in the Oeyama ophiolite

Mean directions were calculated for LTCs and HTCs: the results are shown in Fig. 3 and Table 2. NRM directions with a maximum angular deviation (MAD) <10° were selected for the mean calculation.

(1) **Low temperature component**

A LTC was observed in 37 samples between room temperature and 250 °C prior to the appearance of the HTC (Table 2). The mean direction of the LTC was D = 353.5°, I = 47.0° ($k = 29.2$, $\alpha_{95} = 4.4°$, $n = 37$), this is statistically indistinguishable from the direction of the axial dipole field (D = 0°, I = 52.8°) and the present-day geomagnetic field (International Geomagnetic Reference Field (IGRF): D = 352.3°, I = 50.1°; Thébault et al., 2015).

(2) **High temperature component**

The mean directions of the HTCs were determined based on the data of each category, and from data of integrated categories. As SW remanence was observed in SW_{SW}, SW_{NE}, and SW_{ex} categories, the mean direction for SW remanence in these three categories was D = 226.0°, I = −37.8°, $k = 21.0$, $\alpha_{95} = 5.9°$, $n = 30$. The tilt-corrected directions are listed in Table 2 using the tilt of the early Miocene Tottori Group (N37°W/10E°) or Miocene volcanic rocks (S80°W/8°N). They also reveal a clockwise rotated declination.

The mean direction of the NE remanence was D = 59.0°, I = 39.4°, $k = 72.3$, $\alpha_{95} = 14.6°$, $n = 3$, which is antipodal to the SW remanence. The mean directions for the NE and SW remanences pass the reversal test (McFadden and McElhinny, 1990), with a C classification at the 95% confidence level ($\gamma_o = 10.3° < \gamma_c = 18.7°$). By converting the NE remanence direction to the reversed direction, we calculated a mean direction of D = 227.2°, I = −38.0°,

Table 1
Sample categories and their rock magnetic characteristics. (a) Category of samples. The sample exhibits three directions: SW, NE, and N remanences. Samples with SW remanence were further classified into three categories (SW_{SW}, SW_{NE} and SW_{ex}), determined from thermal demagnetization behavior. Samples with NE and N remanences were classified as categories NE and N, respectively. Circled numbers and numbers are site and sample names, respectively. Samples with an asterisk were subjected to AFD: (b) magnetic carriers in samples of the five categories. SD: single-domain, MD: multi-domain, Large letter: main carrier, small letter: auxiliary carrier.

Remanence	Category	Site (GY)	Samples (GY)
(a) Categories of samples			
SW (34)	SW_{SW} (19)	③	33, 34
		⑧	1118, 83, 85*, 86, 87
		⑦	71*, 75*
		②	22, 25, 26, 28*
		⑤	15, 51, 52, 54, 56,
	SW_{NE} (8)	③	32
		⑧	81
		⑦	73*, 74*
		②	12, 21*, 23-1, 27
		⑤	
	SW_{ex} (7)	③	35, 36*,
		⑧	82, 84*, 88*
		⑦	
		②	
		⑤	16, 53*
NE(3)	NE(3)	③	
		⑧	
		⑦	77*, 78*
		②	23-2
		⑤	
N(10)	N(10)	③	13, 31
		⑧	
		⑦	17, 72*, 76*
		②	14, 24
		⑤	55*, 57, 58*

Remanence	Category	Minerals	Domain state
(b) Rock magnetic characteristics of category			
SW	SW_{SW}	Magnetite	SD + MD
	SW_{NE}	Magnetite	SD
		+ Maghemite	MD
	SW_{ex}	Magnetite	MD + SD
NE	NE	Maghemite	MD + SD
N	N	Magnetite	MD + SD

$k = 22.0$, $\alpha_{95} = 5.4°$, $n = 33$. The NRM direction of the HTC in this locality is characterized by a clockwise rotated direction (∼50°) from the south.

The N remanences observed between room temperature and 450 °C in the ThD indicate an in situ mean direction of D = 350.1°, I = 43.1°, $k = 29.6$, $\alpha_{95} = 9.0°$, $n = 10$. This is statistically indistinguishable from the axial dipole field and the present-day geomagnetic field.

3.4. Alternating field demagnetization

Following the demagnetization of the low coercivity component, three distinct directions with SW, NE, and N remanences were isolated similar to those identified in the ThD (Fig. 4 and Table 2). The mean directions for the three remanences were statistically the same as those isolated during ThD.

The reversed polarity directions of the SW remanence were observed in SW_{SW} and SW_{NE} categories. A distinct peak between 5 and 20 mT in the NRM intensity curve was observed for these samples, followed by a concave upward decay curve (Fig. 4). The SW remanence demonstrated high stability in response to the AFD, and more than 15% of the NRM intensity remained at 60 mT.

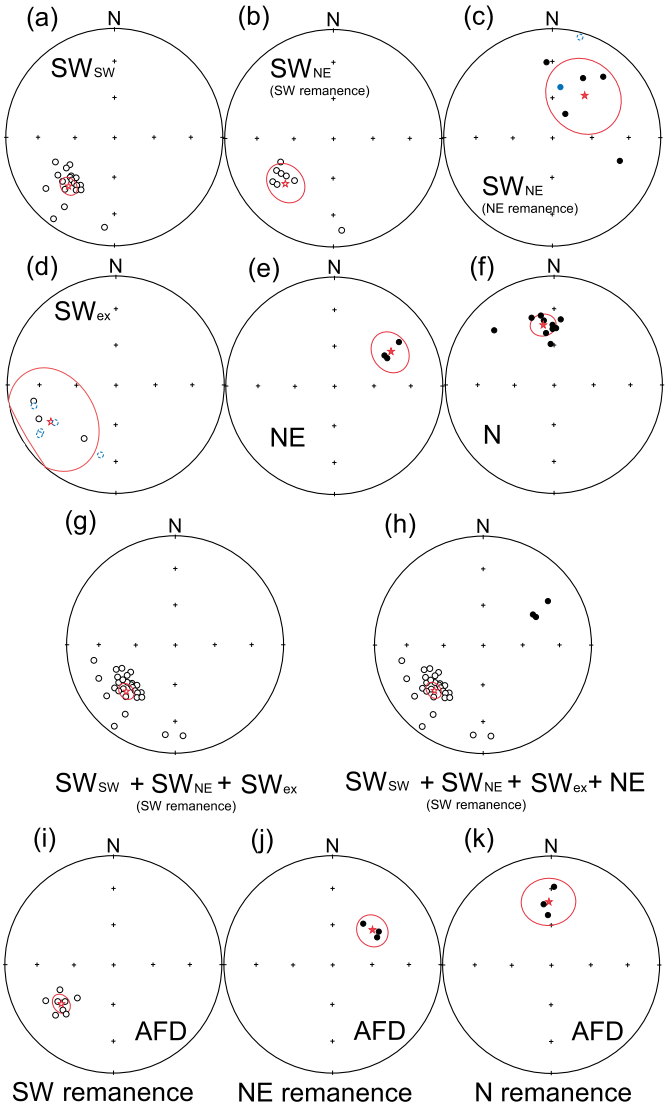


Fig. 3. Equal-area projection of paleomagnetic directions. (a)-(h) Directions after thermal demagnetization for samples of each category and integrated categories. Category: (a) SW_{SW}; (b) and (c) SW_{NE} (SW and NE remanences, respectively); (d) SW_{ex}; (e) NE; and (f) N; (g) Integrated categories of SW_{SW}, SW_{NE} and SW_{ex}; (h) categories SW_{SW}, SW_{NE}, SW_{ex} and NE; (g) and (h); Black circle: data with MAD < 10°; Green circle: data with MAD > 10°; (i)∼(k) Directions after alternating field demagnetization. Remanences: (i) SW; (j) NE; (k) N. Mean directions are calculated from data with MDF > 10° and are denoted by red stars along with ovals indicating the 95% confidence level (see Table 2).

Demagnetization curves for category N revealed a distinct exponential decay, showing low coercivity (Fig. 4). The NRM directions were parallel to the present axial dipole field during AFD between 5 and 35 mT (Fig. 3). A gradual movement of the NRM direction towards the SW was observed above 20 mT in a sample (GY55), suggesting that the SW remanence resides in higher coercivity.

Category SW_{ex} samples exhibited behavior similar to those of samples with SW or N remanences (Fig. 4). Almost all samples showed swift and exponential decay curves in intensity, and their NRM directions were resultant vectors between the SW and N remanences. One sample (GY36) exhibited a SW remanence direction and a distinct peak in intensity at 10 mT, after the NRM intensity swiftly decayed.

The demagnetization curves for category NE samples (GY76-1, GY77, and GY78) were weakly sigmoidal, although the median destructive field (MDF) was less than 8 mT. The NE remanence was

Table 2
Mean direction based on data on samples from each category and from combined categories. n/N: number of samples with MAD < 10°/number of all samples. (inverted): normal direction is inverted to reversed direction. D and I refer to declination and inclination, respectively. k is the Fisherian precision parameter and α_{95} is the radius of the cone at the 95% confidence level with respect to the mean direction (Fisher, 1953). Polarity (N/R): number of samples with normal (reversed) polarity. VGP: virtual geomagnetic pole. A95: the radius of the cone at the 95% confidence level with respect to the VGP. MDF is the median destructive field.

Remanence	Category	n/N	D (°)	I (°)	A_{95} (°)	k	Polarity (N/R)	VGP Latitude	VGP Longitude	
(1) ThD results										
SW	SW_{SW}	19/19	223.4	−38.1	6.6	26.7	(0/19)			
SW	SW_{NE}	8/8	226.6	−39.0	14.6	15.3	(0/8)			
NE	SW_{NE}	6/8	37.3	49.2	28.6	6.4	(6/0)			
SW	SW_{ex}	3/7	240.4	−31.6	36.6	12.4	(0/3)			
NE	NE	3/3	59.1	39.4	14.6	72.3	(3/0)			
SW	SW_{SW} + SW_{NE}	27/27	224.3	−38.3	5.9	22.9	(0/27)			
SW	SW_{SW} + SW_{NE} + SW_{ex}	30/34	226.0	−37.8	5.9	21.0	(0/30)	47.4°N	230.3°E	$A_{95} = 5.7°$
(Tilt corrected)	N37°W/10E°		226.8	−28.2				43.6°N	238.1°E	$A_{95} = 5.7°$
(Tilt corrected)	N80°E/8°N		221.4	−33.1				49.6°N	237.9°E	$A_{95} = 5.7°$
SW	SW_{SW} + SW_{NE} + SW_{ex} + NE(inverted)	33/37	227.2	−38.0	5.5	22.0	(3/30)			
N	N	10/10	350.1	43.1	9.0	29.6	(10/0)			
IGRF			352.3	50.1						

Remanence	n/N	D (°)	I (°)	A_{95} (°)	k	Polarity (N/R)	MDF (mT)
(2) AFD results							
SW	8/8	233.8	−39.5	6.6	70.8	(0/8)	45.4 ± 10.6
NE	3/3	49.1	50.2	11.6	114.9	(3/0)	6.8 ± 2.7
N	3/3	358.0	40.8	19.0	43.2	(3/0)	4.9 ± 1.4

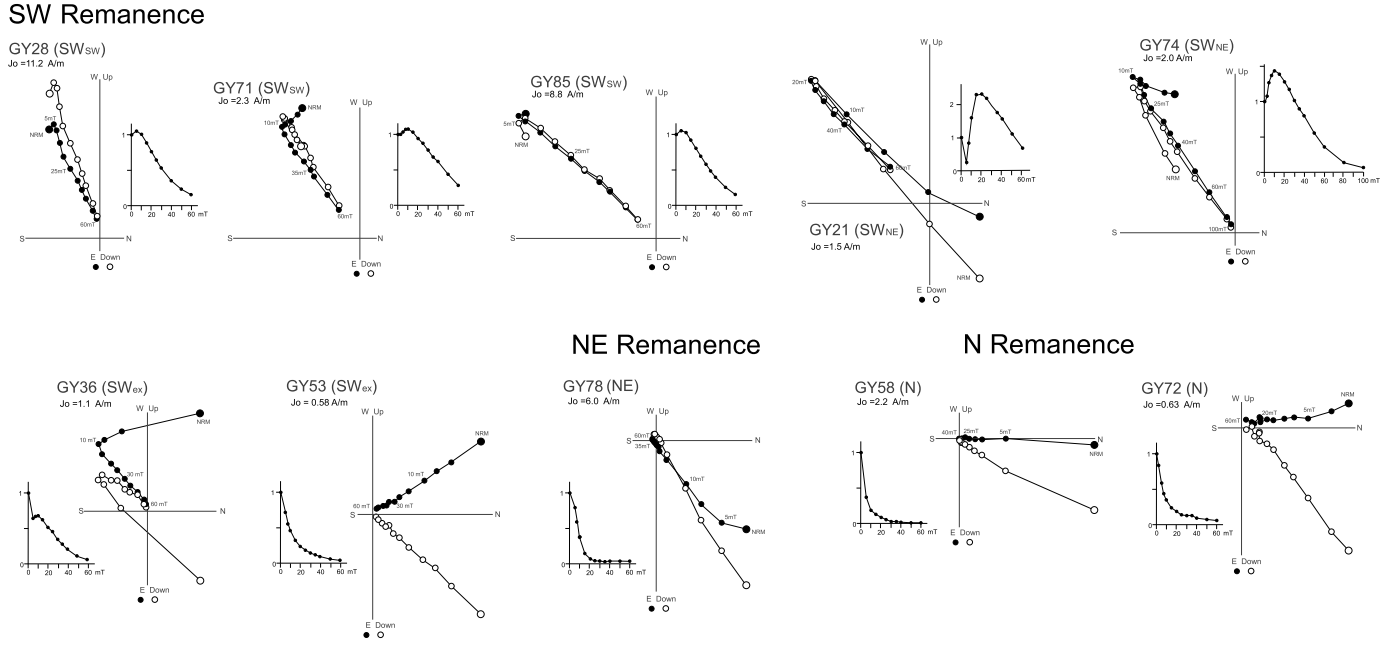


Fig. 4. Zijderveld plots and normalized demagnetization curves during stepwise alternating field demagnetization for samples with the SW, NE and N remanences defined by the thermal demagnetization. Categories of thermal demagnetization behavior are attached to sample numbers. Open (solid) symbols in the Zijderveld plot indicate projection onto a vertical (horizontal) plane. All directions are plotted in geographic coordinates.

definitely observed between 0 and 15 mT. However, following the demagnetization of NE remanence, the SW remanence emerged in the high-field AFD. The NRM direction of GY76-1 and GY78 moves towards the SW above 15 mT. The NE remanences (D ∼ 40°, I ∼ 50°) were also observed in the category SW_{NE} samples (GY21 and GY74) between 0 and 15 mT before the SW remanence was isolated above 20 mT.

4. Degree of serpentinization and microscopic observation

The degree of serpentinization (S) was calculated from bulk density (d) using the equation formulated by Miller et al. (1997), $S = (3.3 - d)/0.785$. The bulk density of a sample was obtained using the buoyancy method from mass and volume measurements were made using an electronic force balance ($\pm 10^{-3}$ g). The degree of serpentinization for each sample is listed in the Appendix Table.

Sample densities in the sampling locality ranged between 2.41–3.21 g/cm^3, corresponding to a range between 99% and 12% in serpentinization degree values. Sample GY72 demonstrated an extraordinarily large serpentinization degree of 114%. Serpentinization was heterogeneous at each site and sample. The serpentinization degree varied from 19% to 114% within sampling site ⑦ and from 30% to 59% for sample GY87.

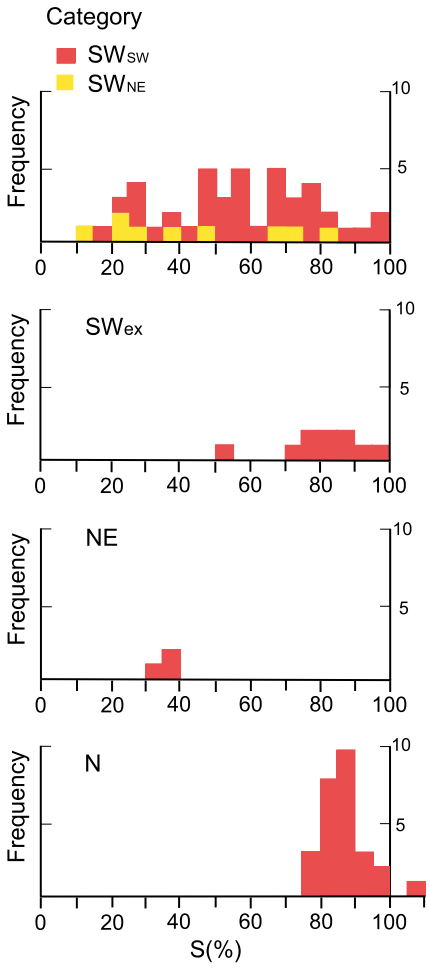


Fig. 5. Frequency distribution of the degree of serpentinization (S). Frequency distribution is depicted for five categories.

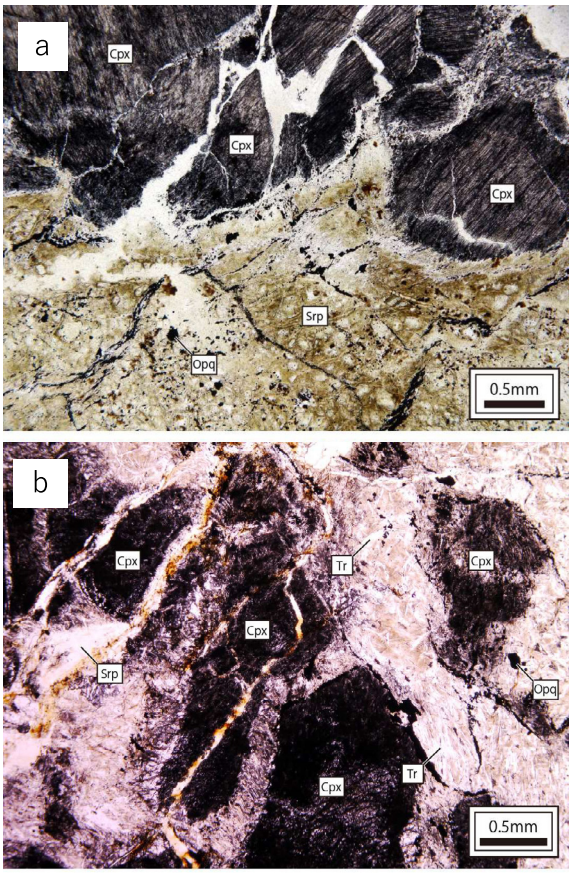


Fig. 6. Optical photo micrographs of open images. (a) Image for GY71 with a relatively low serpentinization degree (56%). Serpentine and tremolite occurred between the crystal grains of coarse and densely packed pyroxene; (b) image for GY73 with a relatively high serpentinization degree (75%). Coarse pyroxene crystals surrounded by fine aggregates of tremolite and serpentinite. Cpx: clinopyroxene, Opq: opaque (magnetite), Srp: serpentine, Tr: tremolite.

Frequency distributions of the degree of serpentinization are shown for each category in Fig. 5. Category N samples showed the strongest degrees of serpentinization of >75%, and a degree >70% is indicated for category SW_{ex} samples with the exception of one data point (GY82; 51%). Category NE samples showed a weak degree of 31–39%. Although the degree was distributed over a wide range between 12% and 99% for 27 samples with the SW_{SW} and SW_{NE} categories, two-thirds of samples (21 samples) showed lower degrees of <66%.

Difference in microstructures of serpentinite was recognized by the degree of serpentinization (Fig. 6). Microstructures were observed under transmitted and reflected light for eight serpentinite samples from site ⑦ of unit B in the outcrop. The textures of samples with fairly low serpentinization degrees (GY71, GY74, GY75, GY77, and GY78; 19–56%) were characterized by the occurrence of serpentine between coarse, densely packed clinopyroxene crystals. Coarse magnetite grains (100–500 μm) was observed between grains of clinopyroxene and olivine, while fine magnetite grains were present along the cleavage in clinopyroxene and olivine. On the other hand, the textures of samples with a greater degree of serpentinization (GY72, GY73, and GY76; 75 ∼ 114%) were characterized by coarse clinopyroxene crystals surrounded by fine aggregates of serpentinite. Coarse magnetite grains up to 500 μm was sparsely distributed, and fine magnetite grains occurred along cracks or formed veins. There was no maghemite identified in any of the eight samples.

5. Rock magnetism

5.1. Thermomagnetic analysis

Thermomagnetic analyses (high-field magnetization as a function of temperature) were conducted on one to three samples across five categories using a laboratory-made magnetic balance at Kyoto University (Fig. 7). Samples were heated and cooled at 8 °C/min in air with a direct current (DC) field between 110–250 mT. The Curie temperatures (T_c) were determined using the intersecting tangents method (Prévot et al., 1983).

All samples had a T_c between 553 and 582 °C, suggesting the presence of magnetite or partial to completely maghemitized magnetite. Magnetite was expected from a smooth decrease in magnetization with increasing temperature up to Tc, with the exception of three samples (GY24, GY36, and GY77).

Category NE sample GY77 exhibited an inflection point at approximately 350 °C, while an inflection point at approximately 500 °C was observed in samples GY36 (category SW_{ex}) and GY24 (category N). A small tail above 580 °C was observed in GY78 (category NE) and GY21 (category SW_{NE}). These features are possibly due to the presence of maghemitized magnetite.

Lower magnetization during cooling than heating was observed for all samples (Fig. 7), indicating that oxidation and/or breakdown of the original magnetic grains occurred during heating in air.

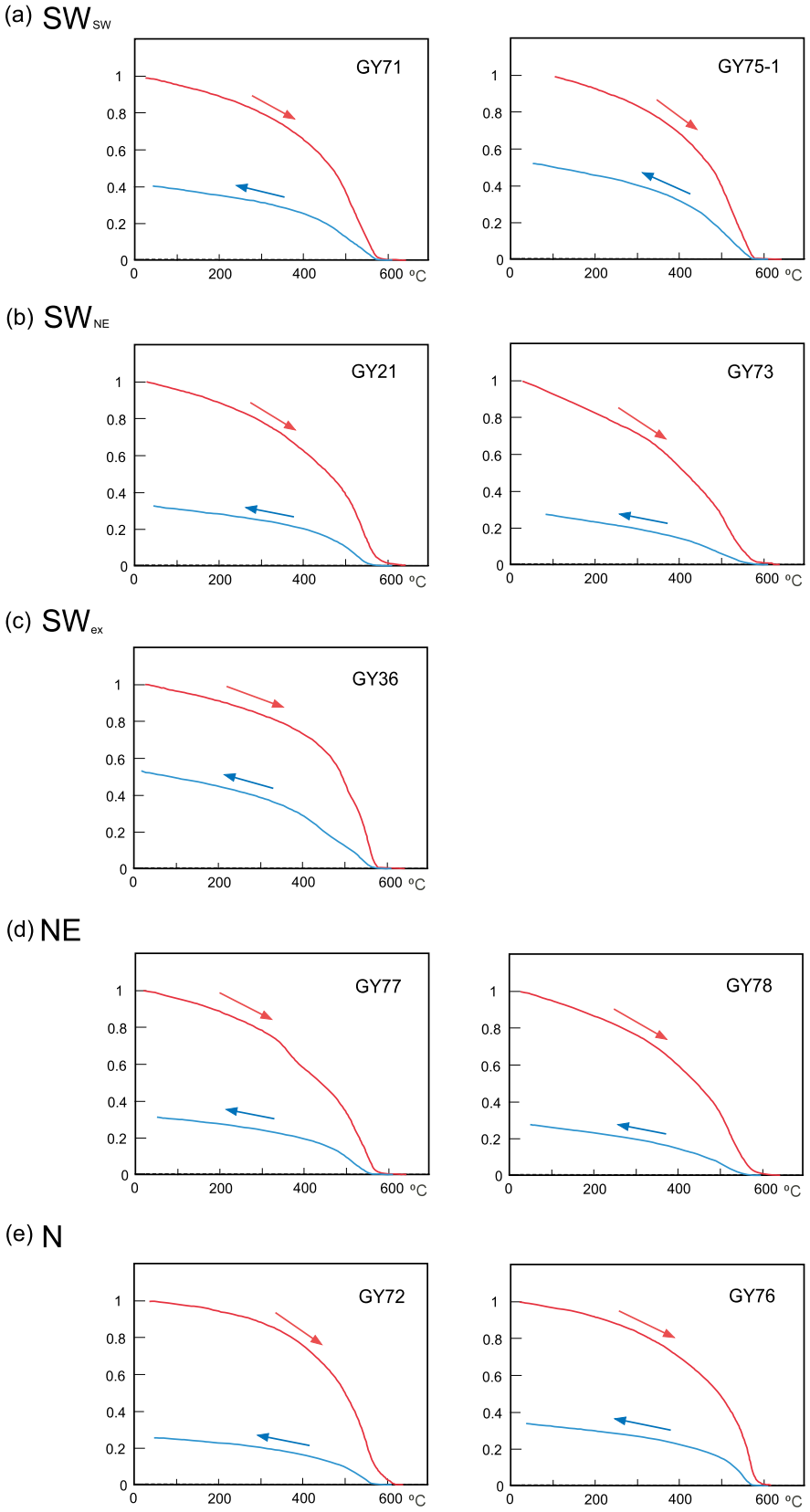


Fig. 7. Strong field thermomagnetic analysis for samples of each category in air condition. Red and blue lines indicate the heating and cooling curves, respectively.

5.2. Median destructive field of alternating field demagnetization

AFD provides the MDF, which supplies information on the magnetic coercivity of samples. As three remanences (SW, NE, and N remanences) appeared during the AFD, the MDF was examined for each remanence (Table 2).

The MDF for SW remanence was measured from SW_{SW} and SW_{NE} category samples. Their values were larger than 30 mT and

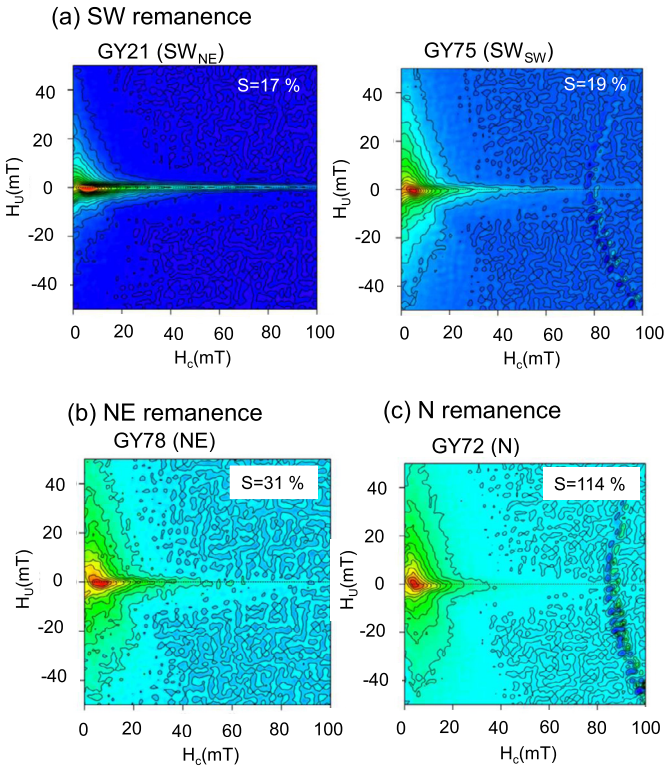


Fig. 8. First-order reversal curves (FORC) diagrams (SF = 3) for samples with SW, NE, and N remanences (Harrison and Feinberg, 2008). VARIFORC was not used and first-point was removed during creating of the FORC diagrams. SW remanence; GY21 of category SW_{NE} and GY75 of category SW_{SW}, NE remanence; GY78 of category NE, and N remanence; GY72 of category N. S is the degree of serpentinization.

the MDF of GY75 was as large as 59 mT: The mean MDF is 45.4 ± 10.6 mT. The N remanence in category N samples had the lowest MDFs, less than 6.5 mT: The mean MDF is 4.9 ± 1.4 mT. As magnetite is a magnetic carrier for samples with SW and N remanences deduced from thermo-magnetic analysis, the MDF is an inverse proxy for magnetic grain size. SD grains (30 mT < MDF) were predominant in samples with the SW remanence, while MD grains (MDF < 10 mT) dominated samples with N remanence.

The MDFs of the NE remanence were less than 8.5 mT, while the mean MDF was 6.8 ± 2.7 mT. MD-like AFD curves were also observed.

5.3. First-Oder Reversal Curves (FORCs)

First-order reversal curve (FORC) experiments for seven samples were carried out using the Princeton Measurements MicroMag model 3900 vibrating sample magnetometer (VSM) of the Paleo and Rock magnetism Laboratory at the Kochi Core Center, Kochi University. During FORC measurements, the number of FROC curves, field step size and measurement time per point were 200, 1.3 mT and 100 ms, respectively. FORC diagrams were created using FORCinel (Harrison and Feinberg, 2008) (Fig. 8).

The FORC distribution of sample GY21 (category SW_{NE}) with a SW remanence indicates dominant fine-grained SD magnetite. The presence of well-dispersed SD grains was confirmed by a strong "central ridge" distribution along the H_c axis up to ~100 mT with no vertical spread (Roberts et al., 2014; Harrison et al., 2018). In samples GY75 (category SW_{SW}) and GY78 (category NE), the central ridge along the H_c axis had decrease and vertical spread along the H_u axis had risen, suggesting increase in the proportion of coarse-grained MD with respect to SD grains (Harrison et al., 2018).

The FORC diagram of a sample (GY72; category N) with N remanence is indicative of a feature for predominantly coarse-grained MD (Pike et al., 2001). The diagram shows a considerable vertical spread (along the H_u axis). There is also a subtle central-ridge-like distribution along the H_c axis ($H_u = 0$), extending to ~ 40 mT. The FORC distribution function abruptly decreased with increasing H_c.

5.4. Day plot

The hysteresis parameters were also measured using the same instrument used for FORC. The hysteresis parameters for eight samples were represented in a Day plot (Fig. 9), with saturation remanent magnetization/saturation remanence (Mrs/Ms) versus remanent coercive force/coercive force (Hcr/Hc), to obtain information on the magnetic domain state and magnetic grain size.

The M_{rs}/M_s measured values for all samples ranged from 0.05 to 0.16, while the ratio of H_{cr}/H_c ranged from 1.6 to 3.7. In terms of hysteresis ratios of M_{rs}/M_s versus H_{cr}/H_c, almost all samples fell in the pseudo-single-domain (PSD) region (Day et al., 1977). A variable proportion of SD and MD grains was anticipated, as data are distributed along the theoretical SD + MD mixing curve of magnetite (Dunlop, 2002). SW remanence samples seem to have a larger SD fraction with respect the MD fraction than the N remanence samples.

5.5. Modified Lowrie-Fuller test

The modified Lowrie-Fuller test (Lowrie and Fuller, 1971) was applied to three samples with a SW or a N remanence, respectively (Fig. 9). An anhysteretic remanent magnetization (ARM) was imparted in a 100 μT biasing DC field with a peak AF of 100 mT, while IRM was produced in a 100 mT field.

SW remanence; GY71, GY75, GY85 (category SW_{SW})

The degrees of serpentinization for three samples vary between 19 and 56%.

The ARM demagnetization curve was considerably more resistant than the IRM curve throughout AFD. The presence of a significant amount of SD-sized grains was demonstrated by the Lowrie-Fuller test (Bailey and Dunlop, 1983; Halgedahl, 1998). Firstly, the MDF of ARM was more than twice that of IRM. Secondly, the ARM decay curve was initially convex upward and then convex downward, which is uncharacteristic of large MD behavior (Dunlop and Ozdemir, 1997). However, the presence of a small amount of MD grains was also expected from the exponential decay curves for IRM. This AFD behavior has been interpreted as being 'bimodal' with SD and MD grains in the grain size distribution, as postulated by Dunlop (1983).

N remanence; GY58, 72, 76 (category N)

The degrees of serpentinization for three samples vary between 75 and 114%.

The IRM demagnetization curve for sample GY76 was similar to the ARM curve in shape and the MDF of the IRM was larger than the MDF of the ARM, indicating the presence of large MD-size grains. Samples GY58 and GY72 showed exponential-like demagnetization curves for the ARM accompanied by a low-coercivity IRM with exponential decay. The AFD behavior for these samples has been interpreted as being 'bimodal' in the grain size distribution (Dunlop, 1983), that is, SD and MD grains. The MDF of ARM does not exceed twice that of IRM, suggesting that the MD fraction is larger than of that a sample with the SW remanence.

6. Discussion

The rock magnetism and paleomagnetism provide information on magnetic carriers and magnetic grain size. In this study,

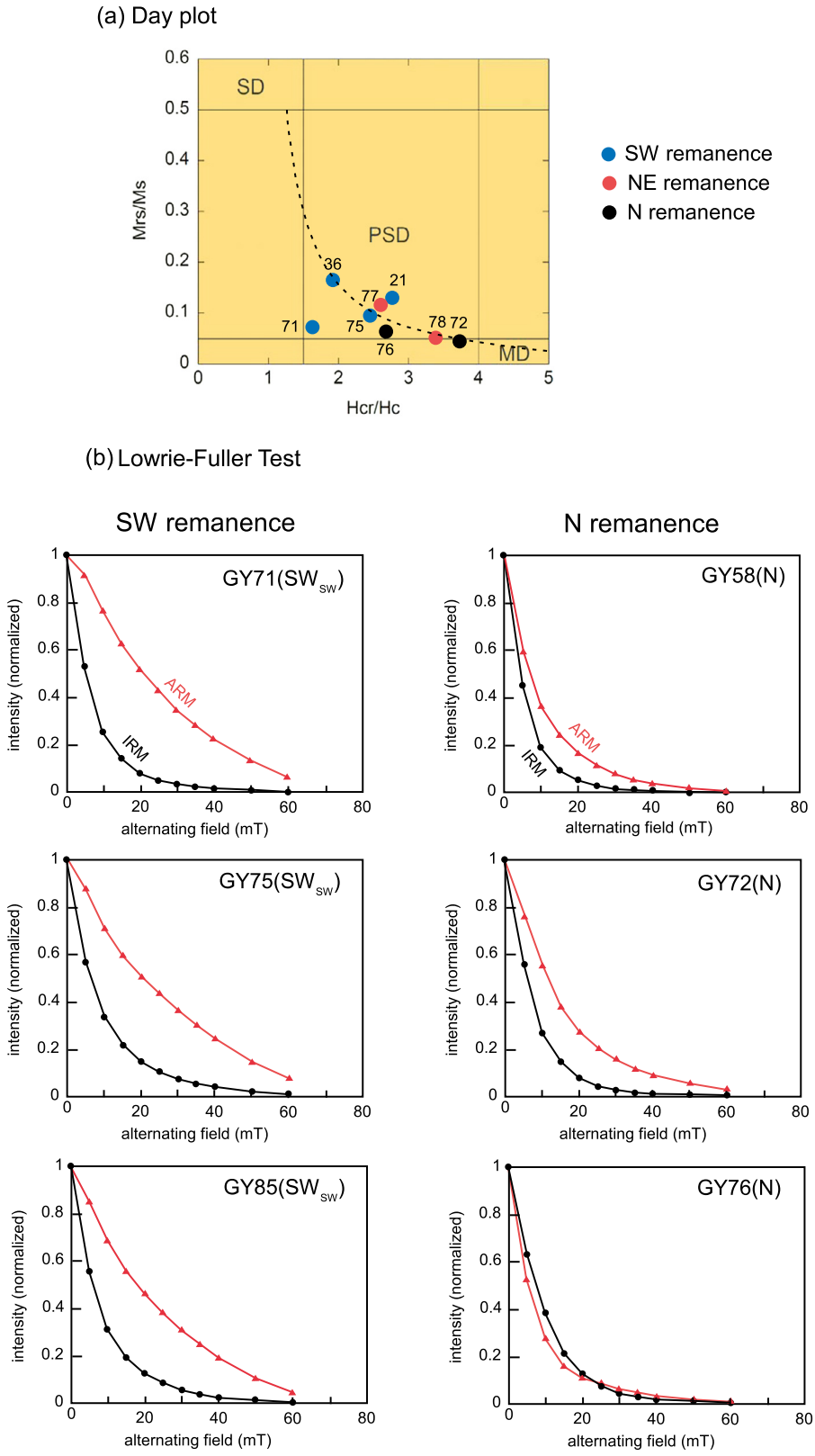


Fig. 9. Rock magnetic results. (a) Hysteresis data for samples with the SW, NE, and N remanences. M_{rs}, saturation remanent magnetization; M_s, saturation magnetization; H_{cr}, remanent coercive force; H_c, coercive force. Single-domain (SD), pseudo-single-domain (PSD) and multi-domain (MD) fields after Day et al. (1977). Almost all samples fall in the PSD region. The dotted curve represents a SD–MD theoretical mixing curve for magnetite (Dunlop, 2002); (b) modified Lowrie-Fuller test: SW remanence (category SW_{SW} GY71, 75, 85) versus N remanence (category N; GY58, 72, 76). The IRM (black line with circles) was produced by applying a 100 mT DC field of and the ARM (red line with triangles) was produced by applying a 100 μT DC biasing field in a peak 100 mT AC field.

we propose an evolutionary aspect of magnetic carriers in the Oeyama ophiolite, and then estimate the timing of serpentinization based on the paleomagnetic direction of the magnetic carrier, which records the Earth's magnetic field during serpentinization.

6.1. Magnetic carriers of remanent magnetizations

The NRM of the serpentine from the Oeyama ophiolite was carried by magnetite or maghemitized magnetite. The MDF was less than 59 mT for samples of all remanences, suggesting that their magnetic carriers have a coercivity much lower than that of hematite. Almost all samples had T_c between 553 and 582 °C (Fig. 7). These results indicate that magnetite is the predominant magnetic carrier. Possible magnetic carriers in samples with the five categories are listed in Table 1.

SW remanence in category SW_{SW} and SW_{NE} samples is carried by magnetite. This is confirmed by a typical smooth decreasing J_s–T curve during heating (Fig. 7). The MDF analysis, FORC diagrams and Lowrie-Fuller test (Figs. 8 and 9) indicate that the magnetization of category SW_{SW} and SW_{NE} samples was carried by SD magnetite grains. Therefore, the SW remanence originated from magnetite SD grains.

The MDF and FORC diagram (Fig. 8) indicate that magnetization of the category N samples was carried by MD magnetite grains; the results from the Lowrie-Fuller test support this conclusion (Fig. 9). The presence of MD grains was also inferred from the exponentially swift decay of the NRM intensity during the stepwise ThD (Fig. 2), as expected from the swift decay of VRM during ThD (Dunlop and Ozdemir, 2000). The N remanence was ascribed to the magnetization of MD magnetite grains.

Magnetization for category SW_{ex} samples is likely to consist of SD grains with SW remanence and MD grains with N remanence. NRM behavior during the ThD was an intermediate between the SW and N remanences; NRM intensity shows a swift decay at 500–530 °C following exponential decay by 100–250 °C.

Maghemitized magnetite is the most plausible candidate for NE remanence. The presence of maghemitized magnetite was identified in a category NE sample (GY77). Although the T_c of this sample is 579 °C, its J_s–T curve shows an inflection point at approximately 350 °C (Fig. 7). This finding is consistent with the inversion temperatures for presumably fine-grained maghemitized magnetite (e.g., de Boer and Dekkers, 1996; Dunlop and Ozdemir, 1997; Liu et al., 1999), or transition temperatures from less stable maghemite into more stable maghemite (Gendler et al., 2005). The thermomagnetic curve also exhibits a small tail just immediately above 580 °C in GY78, indicating the existence of maghemitized magnetite (Helgason et al., 1992). A small tail above 580 °C was observed in the category SW_{NE} (GY21) sample. During ThD and AFD for this sample (Figs. 2 and 4), the NE remanence had a lower coercivity and a T_{ub} higher than those of magnetite; that is further evidence of maghemitized magnetite carrying the NE remanence (Johnson et al., 1975; Goss, 1988; Dunlop and Ozdemir, 1997). The MDF 6.8 mT would be ascribed to the SD grains of maghemitized magnetite.

6.2. Evolution aspect of magnetic carriers of remanent magnetization

The SD grains of magnetite are likely to be the first occurrence of magnetic grains with stable NRM during serpentinization as suggested by Bina and Henry (1990) and Maffione et al. (2013a). The NRM of the SW_{SW} category is the original remanence during the serpentinization in the Oeyama ophiolite.

Serpentinites form via hydration of olivine-rich ultramafic rocks between 200–400 °C and 0.3–0.5 kbar accompanied by the formation of magnetite (Andreani et al., 2013). Serpentinization may be described in a simple form as (e.g., Evans, 2008; Klein et al., 2013):

$$\text{MgFe olivine} + H_2O \rightarrow \text{MgFe serpentine} + \text{magnetite} + \text{MgFe brucite} + H_2 \quad (1)$$

Magnetite is also formed from secondary minerals, such as, brucite and Fe-bearing serpentine (Bach et al., 2006; Frost et al., 2013; Klein et al., 2014; Schwarzenbach et al., 2016) as follows:

$$\text{FeMg brucite} + SiO_2 \rightarrow \text{FeMg serpentine} + \text{magnetite} + H_2O + H_2 \quad (2)$$

$$\text{FeMg brucite} \rightarrow \text{magnetite} + H_2O + H_2 \quad (3)$$

$$\text{Fe serpentine} \rightarrow \text{magnetite} + SiO_2 + H_2O + H_2. \quad (4)$$

Although the specific reactions are complex, serpentinization produces magnetite.

The Fe released from olivine, brucite and serpentine in ordinary serpentinized peridotites is precipitated in situ as fine discrete magnetite grains. As magnetite crystals grow and the crystal size of superparamagnetic magnetite increases to a stable SD size in a magnetic field, the SD magnetite grains acquire grain-growth crystallization remanent magnetization ($CRM_{magnetite}$).

Two major alteration processes are anticipated for the evolution of SD magnetite during serpentinization in a temperature environment lower than ~300 °C, that is, growing into MD grains or maghemitization. First, as serpentinization progresses, the magnetite crystals grow and the magnetite grain size increases from the SD to the MD stage in grain size (Maffione et al., 2013a). The MDF analysis demonstrates that samples with the N remanence are magnetically softer than those with SW remanence. Based on FORC diagrams and the Lowrie-Fuller test (Figs. 8 and 9b), the proportion of the MD/SD magnetite grain size ratio in the N remanence was larger than that of the SW remanence. As the MD magnetite grains are easily able to acquire the present-day magnetic field as a VRM (Shimizu, 1960; Dunlop and Ozdemir, 1997), the NRM of samples with N remanence are parallel to the present-day geomagnetic field direction. Crystal growth from the SD size magnetite to the MD size magnetite is one form of the evolution of magnetic minerals in serpentine. This estimate is plausible as category N samples were limited to serpentine with a high degree of serpentinization (Fig. 5).

Second, an alternative form of evolution is low-temperature oxidation (maghemitization), that is, $Fe_3O_4 \rightarrow \gamma Fe_2O_3$. Maghemitization takes place in an environment <200 °C, occurring mainly at the crystal surface (Dunlop and Ozdemir, 1997). It should be noted that magnetite associated with serpentinization forms between 150 and 350 °C and its effective temperature is approximately 300 °C (Klein et al., 2009). Therefore, after magnetite forms during serpentinization, maghemitization may occur in an environment with decreasing temperature.

6.3. Origin of NE remanence direction with normal polarity

Maghemitized magnetite in NE and SW_{NE} categories acquired the remanent magnetization of another type of CRM (CRM_{γ}) associated with maghemitization during the post-serpentinization period. During serpentinization, the SD magnetite (categories SW_{SW}, SW_{NE} and SW_{ex}) acquired $CRM_{magnetite}$, parallel to the ambient geomagnetic field, and its direction was D ~ 230°, I ~ −40° (Fig. 3). When the originally produced SD magnetite grains altered to the maghemitized magnetite during the maghemitization, the maghemitized magnetite grains secondarily acquired the CRM_{γ}.

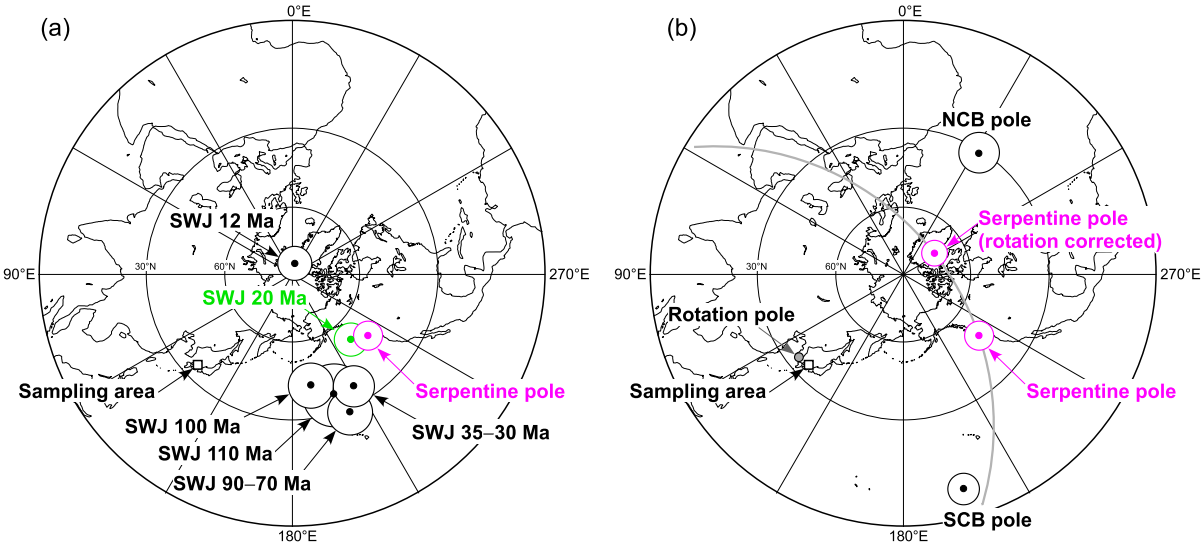


Fig. 10. Stereographic projection of the paleomagnetic pole from the serpentine of the Oeyama ophiolite in the Wakasa area (this study). (a) The serpentine pole position (pink) compared with the APWP of SW Japan (SWJ; 110 Ma to 12 Ma) summarized by Uno et al. (2017); (b) the reconstructed pole position of the serpentine prior to the clockwise rotation of SW Japan (rotation corrected) compared with the Paleozoic pole for NCB (Huang et al., 1999) and SCB (Lin and Fuller, 1990).

We propose two potential factors for the acquisition of the CRM_{γ} with a direction opposite to that of the $CRM_{magnetite}$; ambient geomagnetic field or a self-reversal mechanism. The CRM_{γ} is acquired during maghemitization in the geomagnetic field direction with normal polarity, which occurred following the reversed polarity period. An alternative explanation is the self-reversal phenomenon, in which maghemitized magnetite grows to SD grains and acquires a CRM_{γ} opposite to the direction of the magnetite. The NE remanence directions of the self-reversal origin may be regarded as equivalent for the SW remanence directions of SD magnetite in the SW_{SW}, SW_{NE} and SW_{ex} categories, although its polarity is opposite.

Self-reversal phenomena for titanium-poor magnetite are enigmatic. The self-reversal of titanomagnetites with relatively high Ti content ($x > 0.6$) was discovered in highly oxidized basalts recovered by the Ocean Drilling Program (ODP) (Doubrovine and Tarduno, 2004, 2006; Cavallo et al., 2010). O'Reilly and Banerjee (1966) theoretically proved that the self-reversal phenomenon is restricted to oxidized titanomagnetite. On the other hand, the laboratory simulation of maghemitization oxidized from SD-size titanium-poor magnetites demonstrates that the maghemite acquires the CRM direction parallel to the original NRM direction (Johnson and Merrill, 1972; Ozdemir and Dunlop, 1985).

Although we have proposed either mechanism due to the geomagnetic field or self-reversal for these discovered NE remanence directions with normal polarity, we were unable to determine the responsible origin. Further studies are required to determine the magnetic nature of serpentines.

6.4. When did serpentinization occur?

The timing of serpentinization is estimated on the basis of a comparison between the NRM direction of the Oeyama ophiolite and the paleomagnetic directions in SW Japan. This study proposes two characteristic paleomagnetic directions for serpentinization in the Oeyama ophiolite (Fig. 3). First, the mean SW remanence direction is calculated from the data for categories SW_{SW}, SW_{NE}, and SW_{ex} of the SD grains in the magnetite. The second direction is based on the SW remanence directions of categories SW_{SW}, SW_{NE}, SW_{ex} and NE, assuming that the NE remanence direction is a contemporaneous record of the geomagnetic field in the SW remanence direction. The former mean direction is D = 226.0°, I = −37.8° ($\alpha_{95} = 5.9°$, N = 30), whereas the latter is D = 227.2°, I = −38.0° ($\alpha_{95} = 5.4°$, N = 33). Although the two directions are statistically indistinguishable, we adopted the SW remanence directions as the characteristic direction of serpentinization. This characteristic direction yields a pole position of 47.4°N, 230.3°E with $A_{95} = 5.7°$ for the Oeyama ophiolite.

We confirm that no serpentinization occurred in the Paleozoic when the Oeyama ophiolite occupied the Oeyama area. Because it is controversial whether the SW Japan arc of the Paleozoic era belonged to the NCB or SCB (Isozaki and Maruyama, 1991; Ishiwatari and Tsujimori, 2003), the Oeyama pole at serpentinization is compared with those of the NCB and SCB. Paleomagnetic poles at approximately 450 Ma are constructed from the paleomagnetic data of NCB and SCB (Fig. 10). The paleopole during the Middle Ordovician (472–461 Ma) is estimated at 31.5°N, 327.7°E for the NCB (Huang et al., 1999), while the paleopole during the Silurian (425 Ma) is at 6.8°N, 195.6°E for the SCB (Lin and Fuller, 1990). As SW Japan experienced the clockwise rotation due to opening of the Japan Sea in the early Miocene (Otofuji et al., 1985; Jolivet et al., 1994), the geomagnetic pole for the Oeyama ophiolite must be calculated from the paleoposition before the rotation of SW Japan. SW Japan is reconstructed by counter-clockwise rotation on a Euler pole (34°N, 129°E) by 42° (Otofuji and Matsuda, 1987). Then the geomagnetic pole calculated from the reconstructed position of SW Japan shifts to 73.4°N, 302.0°E. This reconstructed pole position is far from the Paleozoic paleomagnetic poles for the NCB and SCB, suggesting that serpentinization of the Oeyama ophiolite is not ascribable to the Paleozoic phenomenon.

Serpentinization occurred during the Cenozoic. The geomagnetic pole position for the Oeyama ophiolite is compared with the APWP of SW Japan. Uno et al. (2017) reported the APWP of SW Japan between 110 and 12 Ma (Fig. 10). The geomagnetic pole for the Oeyama ophiolite coincides with the 20 Ma pole position (51.3° N, 221.3° E with $A_{95} = 6.7°$) of SW Japan.

The effect of structural correction is examined for the pole position of the Oeyama ophiolite. Using the tilt of the early Miocene Tottori Group and Miocene volcanic rocks, the tilt-corrected paleomagnetic direction yields a pole position at 43.6° N and 238.1°E or 44.0°N and 222.9°E respectively. The tilt-corrected paleomagnetic poles show few large deviations from the pole at 20 Ma of SW Japan and do not approach the 30–110 Ma poles of SW Japan. Therefore, we conclude that the timing of 20 Ma is a plausible

estimate for serpentinization in the Oeyama ophiolite. The serpentinization at the Wakasa area occurred until the opening of the Japan Sea ceased at 15 Ma.

The serpentinization age for the Oeyama ophiolite of 20 Ma is much younger than previously suggested by Tsujimori et al. (2005). A U-Pb age of 472 ± 8.5 Ma was observed for zircon crystals in a jadeite vein in the Osayama serpentine mélange of a part of the Oeyama ophiolite, located to the west of the Wakasa area (Fig. 1). Tsujimori et al. (2005) recognized that this zircon age in jadeite associated with serpentine is the timing of the interaction between alkaline fluid and ultramafic rocks; thus, these data are not direct time constraints of serpentinization. We conclude that the Paleozoic peridotite in the Oeyama ophiolite altered to serpentine at approximately 20 Ma. Alternatively, 20 Ma serpentinization is the most recent phenomenon in multi occurrences of serpentinization since the ophiolite formed. Magnetite originated in older phases of serpentinization grows up to MD grains and finally their NRMs are parallel to the present-day geomagnetic field.

The period of 20 Ma is immediately prior to or during the clockwise rotation of SW Japan accompanied by the opening of the Japan Sea (Otofuji et al., 1991; Jolivet et al., 1994; Baba et al., 2007). Serpentinization at pre or syn-event of the Japan Sea opening is also suggested from a paleomagnetic study of serpentinized ultramafic rocks in the Hayachine tectonic belt, northern Kitakami Terrane, northeast Japan (Otofuji et al., 2003). Although previously serpentinization is considered to have occurred in mid-ocean ridges, seafloor detachment faults or subduction settings (e.g., Coulton et al., 1995; Dubinska et al., 2004; Deschamps et al., 2010; Maffione et al., 2013b), we suggest that serpentinization of the Oeyama ophiolites may be attributed to the opening of a back-arc tectonic setting. The tectonic conditions during the opening of the Japan Sea are likely to have induced fluid migration at low temperatures in rotating SW Japan.

7. Conclusions

Serpentines in the Oeyama ophiolite collected from Wakasa (35.39°N, 134.39°E), SW Japan, provide information on age for serpentinization:

(1) Three directions were identified in the NRM directions of the HTCs: the SW, NE, and N remanences:
(2) Rock magnetism and paleomagnetism show that the SW remanence is carried by SD magnetite grains that originally acquired NRM in the SW remanence direction during serpentinization:
(3) The original SD magnetite experienced two additional evolutionary stages. As serpentinization progressed, the magnetite grains grew to MD-sized grains (N remanence). Alternatively, maghemitization occurred under hydrothermal conditions (NE remanence):
(4) The mean direction is determined for the SW remanence with SD magnetite grains was D = 226.0°, I = −37.8°, $\alpha_{95} = 5.9°$, and $n = 30$, corresponding to the geomagnetic pole at 47.4°N, 230.3°E and $A_{95} = 5.7°$:
(5) The geomagnetic pole position for serpentine with SD magnetite grains overlaps with the 20 Ma pole (51.3°N, 221.3°E with $A_{95} = 6.7°$) in the APWP of SW Japan, indicating that serpentinization occurred at 20 Ma. At a minimum, serpentinization in the Oeyama area occurred during the opening of the Japan Sea.

CRediT authorship contribution statement

Yo-ichiro Otofuji: Conceptualization, Methodology, Investigation, Writing. **Makoto Fujihara:** Methodology, Investigation, Formal analysis **Koji Uno:** Conceptualization, Methodology, Investigation, Formal analysis, Writing

Declaration of competing interest

All authors have no conflicts of interest in academic research.

Acknowledgements

The authors would like to thank Naoto Ishikawa (Kyoto University) for providing the laboratory facilities. We thank Yuhji Yamamoto for FORC and hysteretic parameter measurements at the Paleo and Rock magnetism Laboratory at the Kochi Core Center, Kochi University. We would like to acknowledge the useful suggestions from Masako Miki. We thank K.P. Kodama and M.J. Dekkers for their careful and useful reviews. This research work was partly supported by Grant-in-Aid (No. 23654173) from the Japanese Ministry of Education, Culture, Sports and Technology (MEXT).

Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.epsl.2020.116656.

References

Andreani, M., Munoz, M., Marcaillou, C., Delacour, A., 2013. μXANES study of iron redox state in serpentinization during oceanic serpentinization. Lithos 178, 70–83.
Baba, A.K., Matsuda, T., Itaya, T., Wada, Y., Hori, N., Yokoyama, M., Eto, N., Kamei, R., Zaman, H., Kidane, T., Otofuji, Y., 2007. New age constraints on counter-clockwise rotation of NE Japan. Geophys. J. Int. 171, 1325–1341.
Bach, W., Paulick, H., Garrido, C.J., Ildefonse, B., Meurer, W.P., Humphris, S.E., 2006. Unraveling the sequence of serpentinization reactions: petrography, mineral chemistry, and petrophysics of serpentinites from MAR 15°N (ODP Leg 209, Site 1274). Geophys. Res. Lett. 33. https://doi.org/10.1029/2006GL025681.
Bailey, M.E., Dunlop, D.J., 1983. Alternating field characteristics of pseudo-single-domain (2-14 μm) and multidomain magnetite. Earth Planet. Sci. Lett. 63, 335–352.
Bina, M.M., Henry, B., 1990. Magnetic properties, opaque mineralogy and magnetic anisotropies of serpentinized peridotites from ODP Hole 670A near the Mid-Atlantic Ridge. Phys. Earth Planet. Inter. 65, 88–103.
Cavallo, C., Sainctavit, P., Arrio, M.-A., Guyodo, Y., Penn, R.L., Forsberg, B., Rogalev, A., Wilhelm, F., Smekhova, A., 2010. Self-reversal of magnetization in oceanic submarine basalts studied with XMCD. Geophys. Res. Lett. 37, L11306. https://doi.org/10.1029/2010GL043390.
Coleman, R.G., 1967. Low-temperature reaction zones and Alpine ultramafic rocks of California, Oregon and Washington. U.S. Geol. Surv. Bull. 1247, 1–49.
Coulton, A.J., Haper, G.D., O'Hanley, D.S., 1995. Oceanic versus emplacement age serpentinization in Josephine ophiolite: implication for the nature of the Moho at intermediate and slow spreading ridges. J. Geophys. Res. 100, 22,245–22,260.
Day, R., Fuller, M., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain-size and composition dependence. Phys. Earth Planet. Inter. 13, 260–267.
de Boer, C.B., Dekkers, M.J., 1996. Grain-size dependence of the rock magnetic properties for a natural maghemite. Geophys. Res. Lett. 23, 2815–2818.
Deschamps, F., Guillot, S., Godard, M., Chauvel, C., Andreani, M., Hattori, K., 2010. In situ characterization of serpentinites from forearc mantle wedges: timing of serpentinization and behavior of fluid-mobile elements in subduction zone. Chem. Geol. 269, 262–277.
Doubrovine, P.V., Tarduno, J.A., 2004. Self-reversed magnetization carried by titanomaghemite in oceanic basalts. Earth Planet. Sci. Lett. 222, 959–969.
Doubrovine, P.V., Tarduno, J.A., 2006. Alteration and self-reversal in oceanic basalts. J. Geophys. Res. 111, B12S30. https://doi.org/10.1029/2006JB004468.
Dubinska, E., Bylina, P., Kozlowski, A., Dorr, W., Nejbert, K., Schastok, J., Kulicki, C., 2004. U-Pb dating of serpentinization: hydrothermal zircon from a metasomatic rodingite shell (Sudetic ophiolite, SW Poland). Chem. Geol. 203, 183–203.
Dunlop, D.J., 1983. Determination of domain structure in igneous rocks by alternating field and other methods. Earth Planet. Sci. Lett. 63, 353–367.
Dunlop, D.J., 2002. Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc), 1: theoretical curves and tests using titanomagnetite data. J. Geophys. Res. 107 (B3), 2056. https://doi.org/10.1029/2001JB000486.
Dunlop, D.J., Ozdemir, O., 1997. Rock Magnetism. Cambridge University Press, Cambridge. 573 p.
Dunlop, D.J., Ozdemir, O., 2000. Effect of grain size domain state on thermal demagnetization tails. Geophys. Res. Lett. 27, 1311–1314.

Evans, B.W., 2008. Control of the products of serpentinization by the $Fe^{2+}Mg_{-1}$ exchange potential of olivine and orthopyroxene. J. Petrol. 49, 1873–1887.
Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc Lond. A 217, 295–305.
Frost, B.R., Evans, K.A., Swapp, S.M., Beard, J.S., Mothersole, F.E., 2013. The process of serpentinization in dunite from New Caledonia. Lithos 178, 24–39.
Furuyama, K., Sawada, Y., Itaya, T., Miyake, Y., Kotaki, A., 1997. K-Ar ages of some volcanic rocks from the Yoka Formation, Miocene Hokutan Group, northern Kinki district. Earth Sci. 51, 452–457.
Gendler, T.S., Shcherbakov, V.P., Dekkers, M.J., Gapeev, A.K., Gribov, S.K., McClelland, E., 2005. The lepidocrocite-maghemite-haematite reaction chain – I: acquisition of chemical remanent magnetization by maghemite, its magnetic properties and thermal stability. Geophys. J. Int. 160, 815–832.
Goss, C.J., 1988. Saturation magnetisation, coercivity and lattice parameter changes in the system Fe_3O_4-γFe_2O_3, and their relationship to structure. Phys. Chem. Miner. 16, 164–171.
Halgedahl, S.L., 1998. Revisiting the Lowrie-Fuller test: alternating field demagnetization characteristics of single-domain through multidomain glass-ceramic magnetite. Earth Planet. Sci. Lett. 160, 257–271.
Harrison, R.J., Feinberg, J.M., 2008. FORCinel: an improved algorithm for calculating first-order reversal curve distributions using locally weighted regression smoothing. Geochem. Geophys. Geosyst. 9, Q05016. https://doi.org/10.1029/2008GC001987.
Harrison, R.J., Muraszko, J., Heslop, D., Lascu, I., Muxworthy, A.R., Roberts, A.P., 2018. An improved algorithm for unmixing first-order reversal curve diagrams using principal component analysis. Geochem. Geophys. Geosyst. 19, 1595–1610. https://doi.org/10.1029/2018GC007511.
Helgason, O., Gunnlaugsson, H.P., Steinthorsson, S., Morup, S., 1992. High temperature stability of maghemite in partially oxidized basalt lava. Hyperfine Interact. 70, 981–984.
Huang, B.C., Yang, Z., Otofuji, Y., Zhu, R., 1999. Early Paleozoic paleomagnetic poles from the western part of North China Block and their implications. Tectonophysics 308, 377–402.
Ishikawa, N., Suzuki, T., Ishida, S., 2017. Paleomagnetic studies on Miocene sequences of Hokutan and Tottori Groups in Southwest Japan: implications for Middle Miocene rotational movement of Southwest Japan block associated with the Japan Sea opening. In: Itoh, Y. (Ed.), Evolutionary Models of Convergent Margins-Origin of Their Diversity. INTECH, Rijeka, Croatia, pp. 155–176.
Ishiwatari, A., 1991. Time-space distribution and petrologic diversity of Japanese ophiolites. In: Peters, T.J., Nicolas, A., Coleman, R.G. (Eds.), Ophiolite Genesis and Evolution of the Oceanic Lithosphere. In: Petrology and Structural Geology, vol. 5. Springer, Dordrecht, pp. 723–743.
Ishiwatari, A., 1994. Circum-Pacific Phanerozoic multiple ophiolite belts. In: Ishiwatari, A., Malpas, J., Ishizuka, H. (Eds.), Circum-Pacific Ophiolites (Proceedings of the 29th IGC, Kyoto, Part D). VSP, Utrecht, pp. 7–28.
Ishiwatari, A., Sokolov, S.D., Vysotskiy, S.V., 2003. Petrological diversity and origin of ophiolites in Japan and Far East Russia with emphasis ion deplete harzburgite. Geol. Soc. (Lond.) Spec. Publ. 218, 597–617.
Ishiwatari, A., Tsujimori, T., 2003. Paleozoic ophiolites and blueschists in Japan and Russian Primorye in the tectonic framework of East Asia: a synthesis. Isl. Arc 12, 190–206.
Isozaki, Y., Maruyama, S., 1991. Studies on orogeny based on plate tectonics in Japan and new geotectonic subdivision of Japanese Island. J. Geogr. 100, 697–761.
Johnson, H.P., Merrill, R.T., 1972. Magnetic and mineralogical changes associated with low-temperature oxidation of magnetite. J. Geophys. Res. 77, 334–341.
Johnson, H.P., Lowrie, W., Kent, D.V., 1975. Stability of anhysteretic remanent magnetization in fine and coarse magnetite and maghemite particles. Geophys. J. R. Astron. Soc. 41, 1–10.
Jolivet, L., Tamaki, K., Fournier, M., 1994. Japan Sea, opening history and mechanism: a synthesis. J. Geophys. Res. 99, 22,237–22,259.
Kimura, K., Hayasaka, Y., 2019. Zircon U-Pb age and Nd isotope geochemistry of latest Neoproterozoic to early Paleozoic Oeyama ophiolite: evidence for oldest MORB-type oceanic crust in Japanese accretionary system and its tectonic implications. Lithos 342–343, 345–360.
Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. Int. 62, 699–718.
Klein, F., Bach, W., Jons, N., McCollom, T., Moskowitz, B., Berquo, T., 2009. Iron partitioning and hydrogen generation during serpentinization of abyssal peridotite from 15°N on Mid-Atlantic Ridge. Geochim. Cosmochim. Acta 73, 6868–6893.
Klein, F., Bach, W., McCollom, T.M., 2013. Compositional controls on hydrogen generation during serpentinization of ultramafic rocks. Lithos 178, 55–69.
Klein, F., Bach, W., Humphris, S.E., Kahl, W.A., Jöns, N., Moskowitz, B., Berquo, T.S., 2014. Magnetite in seafloor serpentinite—some like it hot. Geology 42, 135–138.
Li, X-P., Zhang, L-F., Wilde, S.A., Song, B., Liu, X-M., 2010. Zircons from rodingite in the Western Tianshan serpentinite complex: mineral chemistry and U-Pb ages define nature and timing of rodingitization. Lithos 118, 17–34.
Lin, J.L., Fuller, M., 1990. Palaeomagnetism, North China and South China collision, and the Tan-Lu fault. Philos. Trans. R. Soc. Lond. A 331, 589–598.
Liu, X.M., Hesse, P., Rolph, T., 1999. Origin of maghemite in Chinese loess deposits: aeolian or pedogenic. Phys. Earth Planet. Inter. 112, 191–201.
Lowrie, W., Fuller, M., 1971. On the alternating field demagnetization characteristics of multidomain thermoremanent magnetization in magnetite. J. Geophys. Res. 76, 6349–6399.
Maffione, M., Morris, A., Plumper, O., van Hinsbergen, D.J.J., 2013a. Magnetic properties of variable serpentinized peridotites and their implication for the evolution of oceanic core complexes. Geochem. Geophys. Geosyst. 15, 923–944. https://doi.org/10.1002/2013GC004993.
Maffione, M., Morris, A., Anderson, M.W., 2013b. Recognizing detachment-mode seafloor spreading in the deep geological past. Sci. Rep. 3, 2336. https://doi.org/10.1038/srep02336.
Matsumoto, T., 1986. Stratigraphy of the Miocene series in the area southeast of Tottori City, Tottori Prefecture, Southwest Japan. J. Geol. Soc. Jpn. 92, 269–287.
Matsumoto, T., 1989. Stratigraphical study of the Miocene series in the eastern part of Tottori Prefecture, Southwest Japan. J. Sci. Hiroshima Univ., Ser. C 9, 199–235.
McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism. Geophys. J. Int. 103, 725–729.
Miller, D.J., Christensen, N.I., 1997. Seismic velocities of lower crustal and upper mantle rocks from the slow spreading mid-Atlantic ridge, south of the Kane transform zone (MARK). In: Karson, J.A., Cannat, M., Miller, D.J., Elthon, D. (Eds.), Proc. Ocean Drill. Program, Sci. Results, vol. 153, pp. 437–454.
Moreno, T., Wallis, S., Kojima, T., Gibbons, W., 2016. The Geology of Japan. The Geological Society, London. 539 pp.
Nishimura, Y., Shibata, K., 1989. Modes of occurrence and K-Ar ages of metamorphic rocks in the 'Sangun metamorphic belt', southwest Japan. Mem. Geol. Soc. Jpn. 33, 343–357.
O'Hanley, D.S., 1996. Serpentinites: Records of Tectonic and Petrologic History. Oxford University Press, Oxford, 277 pp.
O'Reilly, W., Banerjee, S.K., 1966. Oxidation of titanomagnetites and self - reversal. Nature 211, 26–28.
Otofuji, Y., Matsuda, T., Nohda, S., 1985. Opening mode of the Japan Sea inferred from the palaeomagnetism of the Japan Arc. Nature 317, 603–604.
Otofuji, Y., Matsuda, T., 1987. Amount of clockwise rotation of Southwest Japan - fan shape opening of the southwestern part of the Japan Sea. Earth Planet. Sci. Lett. 85, 289–301.
Otofuji, Y., Itaya, T., Matsuda, T., 1991. Rapid rotation of southwest Japan-palaeomagnetism and K-Ar ages of Miocene volcanic rocks of southwest Japan. Geophys. J. Int. 105, 397–405.
Otofuji, Y., Takemoto, K., Zaman, H., Nishimitsu, Y., Wada, Y., 2003. Cenozoic remagnetization of the Paleozoic rocks in the Kitakami massif of northeast Japan, and its tectonic implications. Earth Planet. Sci. Lett. 210, 203–217.
Ozdemir, O., Dunlop, D.J., 1985. An experimental study of chemical remanent magnetizations of synthetic monodomain titanomaghemites with initial thermoremanent magnetizations. J. Geophys. Res. 90, 11513–11523.
Pike, C.P., Roberts, A.P., Dekkers, M.J., Verosub, K.L., 2001. An investigation of multidomain hysteresis mechanisms using FORC diagrams. Phys. Earth Planet. Inter. 126, 11–25.
Prévot, M., Mankinen, E.A., Grommé, S., Lecaille, A., 1983. High paleointensities of the geomagnetic field from thermomagnetic studies on Rift Valley pillow basalts from the Mid-Atlantic Ridge. J. Geophys. Res. 88, 2316–2326.
Roberts, A.P., Heslop, D., Zhao, X., Pike, C.R., 2014. Understanding fine magnetic particle systems through use of first-order reversal curve diagrams. Rev. Geophys. 52, 557–602. https://doi.org/10.1002/2014RG000462.
Schwarzenbach, E.M., Caddick, M.J., Beard, J.S., Bodnar, R.J., 2016. Serpentinization, element transfer, and the progressive development of zoning in veins: evidence from a partially serpentinized harzburgite. Contrib. Mineral. Petrol. 171, 5. https://doi.org/10.1007/s00410-015-1219-3.
Shimizu, Y., 1960. Magnetic viscosity of magnetite. J. Geomagn. Geoelectr. 11, 125–138.
Thébault, E., Finlay, C.C., Beggan, C.D., Alken, P., Aubert, J., Barrois, O., Bertrand, F., Bondar, T., Boness, A., Brocco, L., Canet, E., Chambodut, A., Chulliat, A., Coïsson, P., Civet, F., Du, A., Fournier, A., Fratter, I., Gillet, N., Hamilton, B., Hamoudi, M., Hulot, G., Jager, T., Korte, M., Kuang, W., Lalanne, X., Langlais, B., Léger, J.-M., Lesur, V., Lowes, F.J., Macmillan, S., Mandea, M., Manoj, C., Maus, S., Olsen, N., Petrov, V., Ridley, V., Rother, M., Sabaka, T.J., Saturnino, D., Schachtschneider, R., Siro, O., Tangborn, A., Thomson, A., Tøffner-Clausen, L., Vigneron, P., Wardinski, I., Zvereva, T., 2015. International geomagnetic reference field: the 12th generation. Earth Planets Space 67, 79. https://doi.org/10.1186/s40623-015-0228-9.
Tsujimori, T., Itaya, T., 1999. Blueschist-facies metamorphism during Paleozoic orogeny in southwestern Japan: phengite K-Ar ages of blueschist-facies tectonic blocks in a serpentinite mélange beneath early Paleozoic Oeyama ophiolite. Isl. Arc 8, 190–205.
Tsujimori, T., Nishina, K., Ishiwatari, A., Itaya, T., 2000. 443-403 Ma kyanite-bearing epidote amphibolite from the Fuko Pass metacumulate in the Oeyama area, the Inner Zone of southwestern Japan. J. Geol. Soc. Jpn. 106, 646–649.
Tsujimori, T., Liou, J.G., Wooden, J., Miyamoto, T., 2005. U-Pb Dating of large zircons in low-temperature jadeitite from the Osayama Serpentinite Mélange, Southwest Japan: insights into the timing of serpentinization. Int. Geol. Rev. 47, 1048–1057.
Uemura, F., Sakamoto, T., Yamada, N., 1979. Geology of the Wakasa District. Quadrangle series, scale 1:50,000. Geol. Surv. Jpn., 91 p.

Uno, K., Furukawa, K., Hatanaka, Y., 2017. An analysis of apparent polar wander path for southwest Japan suggests no relative movement with respect to Eurasia during Cretaceous. Phys. Earth Planet. Inter. 267, 19–30.

Uto, K., Tagami, T., Uchiumi, S., 1994. K-Ar and fission-track dating on volcanic rocks of Pliocene Teragi Group from eastern San'in region, Southwest Japan. J. Geol. Soc. Jpn. 100, 787–798.

Wallis, S.R., Yamaoka, K., Mori, H., Ishiwatari, A., Miyazaki, K., Ueda, H., 2020. The basement geology of Japan from A to Z. Isl. Arc 29, e12339. https://doi.org/10.1111/iar.12339.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Method in Palaeomagnetism. Elsevier, Amsterdam, pp. 254–286.